

HeliXplore: A Python package for analyzing multi-strand helix deformations

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Summary

Multi-stranded helices are a recurring structural motif in biomolecular systems, most prominently found in DNA and collagen (Ramachandran & Kartha, 1954; Watson & Crick, 1953). They also appear in synthetic polymers and macromolecules (Yashima et al., 2009). Helix properties depend on strand length, strand composition and the environment, which can be modeled using classical Molecular Dynamics (MD). However, multi-stranded helices can undergo local and global deformations that directly impact their function. MD lacks a tool to systematically quantify deformation across systems and conditions. In the past, people have used polysaccharide atom distances (Khatami et al., 2021), collagen cross-sectional triangles (Ravikumar & Hwang, 2008), pitch, or principal axis measures (Koneru et al., 2019; Zhang et al., 2006). Although adequate for the system under investigation, these metrics are not generalizable and miss key local or collective distortions. Without local or inter-strand descriptors, comparisons across systems remain largely qualitative.

We present HeliXplore, an open-source Python package for the systematic and quantitative analysis of multi-strand helix deformation. Originally inspired by collagen, HeliXplore is generalizable to any helical bundle, including single-stranded helices. HeliXplore measures how helices deviate from their ideal geometry using user-defined backbone atoms. HeliXplore runs calculations in three sections: Section 1 for intra-strand deformations (rise, radius, twist and windowed deviations) per atom, or group of atoms and per time frame and helical regularity per strand; Section 2 for inter-strand deformations (axial shifts, axis angle deviations, axis distance deviations and averaged distance deviations); and Section 3 for triple-helix deformations, using the area and shape of the cross-sectional triangle.

In practice, one only needs HeliXplore.py, the MD trajectory file (in TINKER .arc format or the standard RCSB .pdb format) and the number of strands to be able to run the code. Users can also input the atom names or atom types (for TINKER .arc format) to mark the backbone. `read_tinker_arc()` and `read_traj_pdb()` functions can be replaced to cater to other trajectory file formats. HeliXplore checks for the four required Python dependencies (numpy, scipy, pandas and matplotlib) before running the main code. No other installations are required. For a detailed description of the inputs and examples, see the README file on GitHub. A shorter description is provided with `python HeliXplore.py --help`.

Statement of Need

HeliXplore provides the first open-source, Python-based implementation of a quantitative framework for analyzing multi-helix deformations. By resolving intra- and inter-strand deformations, HeliXplore enables atomic resolution of structural distortions that have previously eluded MD studies. More broadly, HeliXplore establishes a transferable

methodology for analyzing helices, positioning it as a foundational tool for systematic comparisons across systems, conditions and force fields. Users of HeliXplore are free to modify the code and underlying mathematical formulations to adapt to their specific research needs.

Mathematics

Section I:

For a helix of length N , with $\mathbf{p}^{i,m}(t)$ the coordinates of atom i in strand m at time t , the helical axis vector $\mathbf{v}_1^m(t)$ is determined via principal component analysis:

$$\left[\frac{1}{N} \sum_{i=1}^N (\mathbf{p}^{i,m}(t) - \mathbf{c}^m(t))(\mathbf{p}^{i,m}(t) - \mathbf{c}^m(t))^T \right] \mathbf{v}_1^m(t) = \lambda_1^m(t) \mathbf{v}_1^m(t),$$

where $\mathbf{c}^m(t)$ is the average coordinates of strand m at time t :

$$\mathbf{c}^m(t) = \frac{1}{N} \sum_{i=1}^N \mathbf{p}^{i,m}(t),$$

and λ_1^m indicates the principal eigenvalue. The unit vector of $\mathbf{v}_1^m(t)$ is hereafter referred to as $\hat{\mathbf{v}}_1^m(t)$.

Deviations in rise for atom i at strand m are calculated as:

$$\delta_{\text{Rise}}^{i,m}(t) = \frac{\overbrace{(\mathbf{p}^{i+1,m}(t) - \mathbf{p}^{i,m}(t)) \cdot \mathbf{v}_1^m(t)}^{\text{Rise}^{i,m}(t)} - \text{Rise}^{i,m}(0)}{\text{Rise}^{i,m}(0)}.$$

Deviations in radius are calculated using the circumradius $R^{i,m}$ as:

$$\delta_{\text{Radius}}^{i,m}(t) = \frac{R^{i,m}(t) - R^{i,m}(0)}{R^{i,m}(0)},$$

where:

$$R^{i,m}(t) = \frac{\|\mathbf{p}^{i,m}(t) - \mathbf{p}^{i-1,m}(t)\| \|\mathbf{p}^{i+1,m}(t) - \mathbf{p}^{i-1,m}(t)\| \|\mathbf{p}^{i,m}(t) - \mathbf{p}^{i+1,m}(t)\|}{4 \times \frac{1}{2} \|(\mathbf{p}^{i,m}(t) - \mathbf{p}^{i-1,m}(t)) \times (\mathbf{p}^{i+1,m}(t) - \mathbf{p}^{i-1,m}(t))\|}.$$

Deviations in twist for atom i on strand m at time t are calculated as the angle between the normals of the planes defined by consecutive atoms $(i-1, i, i+1)$ ($\mathbf{n}_a^{i,m}(t)$) and $(i, i+1, i+2)$ ($\mathbf{n}_b^{i,m}(t)$) as:

$$\delta_{\text{Twist}}^{i,m}(t) = \frac{\overbrace{\arccos \left(\frac{\mathbf{n}_a^{i,m}(t) \cdot \mathbf{n}_b^{i,m}(t)}{\|\mathbf{n}_a^{i,m}(t)\| \|\mathbf{n}_b^{i,m}(t)\|} \right)}^{\text{Twist}^{i,m}(t)} - \text{Twist}^{i,m}(0)}{\text{Twist}^{i,m}(0)}.$$

Windowed deviations are calculated after superposition using a window of 5 atoms centered around atom i with the Kabsch algorithm (Kabsch, 1976) as:

$$d_{\text{windowed}}^{i,m}(t) = \|\mathbf{q}^{i,m}(t) - \mathbf{p}^{i,m}(0)\|,$$

61 where $\mathbf{q}^{i,m}(t)$ are the aligned coordinates of atom i . $\mathbf{q}^{i,m}(t)$ is calculated from the coordinates
62 of the atoms within the window $\mathbf{p}^{j,m}(t)$, where $j \in [i-2, i+2]$ as:

$$\mathbf{q}^{j,m}(t) = \mathbf{R}^{j,m}(t) (\mathbf{p}^{j,m}(t) - \langle \mathbf{p}^{j,m}(t) \rangle_j) + \langle \mathbf{p}^{j,m}(0) \rangle_j,$$

63 where the optimal rotation matrix $\mathbf{R}^{j,m}(t)$ is obtained by singular value decomposition (using
64 `NUMPY.LINALG.SVD`) of the cross-covariance matrix $\mathbf{H}^{j,m}(t)$:

$$\mathbf{H}^{j,m}(t) = \sum_j (\mathbf{p}^{j,m}(t) - \langle \mathbf{p}^{j,m}(t) \rangle_j) (\mathbf{p}^{j,m}(0) - \langle \mathbf{p}^{j,m}(0) \rangle_j)^T.$$

65 **Helical regularity** for a strand m is calculated as:

$$\mathcal{R}^m(t) = \frac{\sigma_{\text{Rise}}(t)}{\langle |\text{Rise}^{i,m}(t)| \rangle_i} + \frac{\sigma_{\text{Radius}}(t)}{\langle |\text{Radius}^{i,m}(t)| \rangle_i} + \frac{\sigma_{\text{Twist}}(t)}{\langle |\text{Twist}^{i,m}(t)| \rangle_i},$$

66 where σ is the standard deviation of the metric over i .

67 Section II:

68 **Axial shift** of strand m with respect to strand n along the central axis is calculated as:

$$s^{mn}(t) = |\langle \tilde{\mathbf{v}}_1^m(t) \rangle_m \cdot (\mathbf{c}^m(t) - \mathbf{c}^n(t))|, \quad \text{where} \quad \tilde{\mathbf{v}}_1^n(t) = \begin{cases} -\hat{\mathbf{v}}_1^n, & \text{when } \hat{\mathbf{v}}_1^m \cdot \hat{\mathbf{v}}_1^n < 0 \\ \hat{\mathbf{v}}_1^n, & \text{otherwise.} \end{cases}$$

69 **Deviations in axis angles** are calculated between the unit vectors of the principal axes of the
70 two strands m and n as:

$$\delta_{\theta}^{mn}(t) = \frac{\theta^{mn}(t)}{|\arccos(\hat{\mathbf{v}}_1^m(t) \cdot \hat{\mathbf{v}}_1^n(t)) - \theta^{mn}(0)|}.$$

71 **Deviations in axis distances** are calculated using the perpendicular distance between the axes
72 of strands m and n as:

$$\delta_d^{mn}(t) = \frac{d^{mn}(t)}{|\overbrace{(\mathbf{c}^m(t) - \mathbf{c}^n(t)) \cdot (\hat{\mathbf{v}}_1^m(t) \times \hat{\mathbf{v}}_1^n(t))}^{d^{mn}(t)} - d^{mn}(0)|}.$$

73 **Deviations in averaged distances** are calculated using the distance between the average
74 coordinates of strands m and n as:

$$\delta_c^{mn}(t) = \frac{c^{mn}(t)}{|\overbrace{\|\mathbf{c}^m(t) - \mathbf{c}^n(t)\|}^{c^{mn}(t)} - c^{mn}(0)|}.$$

75 Section III:

76 For triple helices, one atom i on each strand is taken to form a triangular cross-section.

77 Deviations in area are calculated from the area of the triangle as:

$$\delta_{\text{Area}}^i(t) = \frac{\frac{1}{2} \overbrace{\|(\mathbf{p}^{i,2}(t) - \mathbf{p}^{i,1}(t)) \times (\mathbf{p}^{i,3}(t) - \mathbf{p}^{i,1}(t))\|}^{\text{Area}^i(t)} - \text{Area}^i(0)}{\text{Area}^i(0)}.$$

78 Deviations in shape are calculated from the normalized isoperimetric ratio (IP) as:

$$\delta_{\text{Shape}}^i(t) = \frac{\frac{\overbrace{\text{IP}^i(t)}^{4\pi \text{Area}^i(t)}}{P^i(t)^2} - \text{IP}^i(0)}{\text{IP}^i(0)},$$

79 where $P^i(t)$ is the corresponding perimeter.

80 The first frame is taken as the reference. The reference can be changed by appending a new
81 reference frame to the beginning of the input MD trajectory. All mathematical details are also
82 outlined as comments in the code.

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